Benjamin Haase

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Education

University of Washington Bothell Bothell, WA
Bachelor of Science in Electrical Engineering | GPA: 3.98/4.0 *Expected June 2028*

Relevant Coursework: Power Electronics (In Progress), Analog Circuits (In Progress)

Technical Skills

Hardware & PCB: Altium Designer, KiCad, Schematic Capture, PCB Layout
Embedded Systems: Bare-metal C, STM32 Microcontrollers, OpenOCD, GDB, SPI, I2C
Software & Tools: LTspice, Git/Github, Power Supplies, Oscilloscopes, Digital Multimeters, Function Generators
Familiar with: Matlab, Python, Verilog

Experience

Shell Eco-marathon EV Design Team (UWB Supermileage) Bothell, WA
Electrical Hardware Engineer *January 2026 – Present*

- Designed and routed a custom printed circuit board using Altium Designer to consolidate the vehicle's auxiliary functions on a single board, allowing the driver to control the lights, horn, windshield wiper, and observe the vehicle's speed.
- Designed and tested a passive RC filter to integrate a Hall-effect sensor with an STM32F446RET6 timer channel to compute vehicle speed in real time.
- Developed bare-metal C firmware using an interrupt-driven timer and blocking SPI communication to calculate and display vehicle speed in real time via a MAX7221CWG LED driver.
- Engineered a high-voltage battery disconnect and safety isolation system utilizing a TE Connectivity Kilovac EV200 contactor to ensure strict competition compliance.

Personal Projects

Bass Guitar Tremolo Effect Pedal | *Mixed-Signal PCB, Altium, LTspice, C*

- Designed and simulated a discrete analog voltage-controlled amplifier (VCA) and a passive peak finder circuit in LTspice to achieve a dynamic tremolo effect with a 40dB modulation depth.
- Routed a mixed-signal PCB in Altium Designer, strictly partitioning analog and digital domains to mitigate digital switching noise and preserve audio signal integrity.
- Developed bare-metal C firmware using a timer-driven ADC via SPI to sample the peak voltage at a calculated frequency, ensuring the modulation rate remained highly responsive to the musician's playing dynamics.